

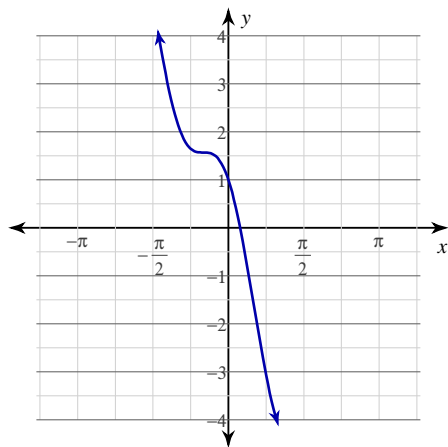
Newton's Method

Date _____ Period _____

Use two iterations of Newton's Method to approximate the real zeros of each function. Use the provided initial guess.

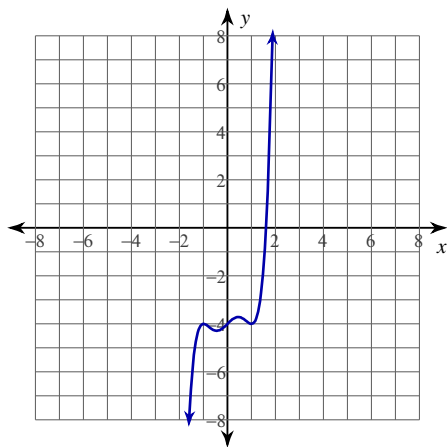
1) $y = \cos 3x - 3x$

Guess: 0.4



2) $y = x^5 - 2x^3 + x - 4$

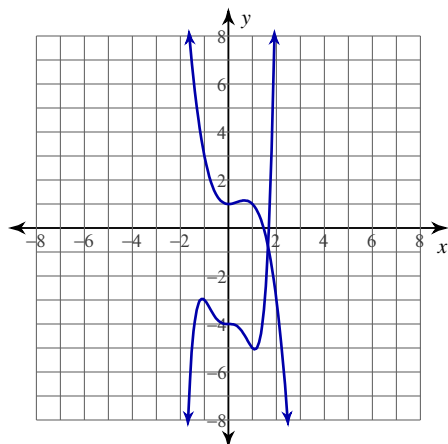
Guess: 1.8



Use Newton's Method to approximate the x -coordinates where the two functions intersect.

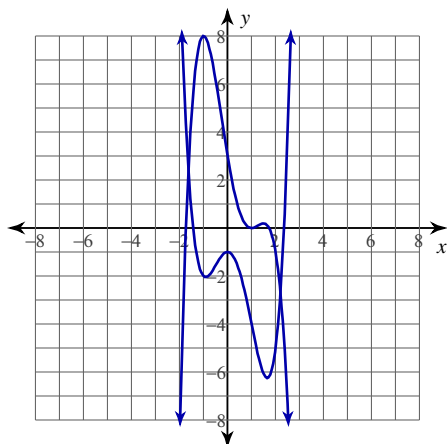
3) $y = -x^3 + x^2 + 1$

$y = x^5 - 2x^3 - 4$



4) $y = x^4 - x^3 - 3x^2 - 1$

$y = -x^4 + 2x^3 + 2x^2 - 6x + 3$



For each problem, use Newton's Method to approximate the positive root.

5) $\sqrt[5]{4}$

6) $\sqrt[5]{7}$